where $\lambda_{(1-\delta)U}\left(e^{W^V}\right) \vee \nu_{(1-\delta)V}\left(e^{W_U}\right) < \infty$ by Condition $\mathcal{H}_\delta(U,V)$. This ends the proof of (D.12).

Using (D.10) we see that, for any $n \geqslant 1$,

$$-\mathcal{W}_{V,U} - W^V + \nu_V(V) \leqslant U_{2(n+1)} - U_{2n} \leqslant \mathcal{W}_{V,U} + W^V - \nu_V(V)$$

as well as

$$-\mathcal{W}^{U,V} - W_U + \nu_V(V) \leqslant V_{2(n+1)} - V_{2n} \leqslant \mathcal{W}^{U,V} + W_U - \nu_V(V).$$

Then, recalling that $U_{2n+1} = U_{2n}$ and $V_{2(n+1)} = V_{2n+1}$ and using Eqs. (E.1) and (E.2) in the proof of the commutation formulae (D.1), we readily obtain the estimates

$$e^{\nu_V(V)}\|e^{\mathcal{W}^{U,V}}\|^{-1}_{\mathbb{L}_\infty(\lambda)}\; e^{-W_U} \leqslant \frac{d\pi_{2n}}{d\nu_V} \leqslant e^{-\nu_V(V)}\|e^{\mathcal{W}^{U,V}}\|_{\mathbb{L}_\infty(\lambda)}\; e^{W_U}$$

as well as

$$e^{\nu_V(V)}\|e^{\mathcal{W}_{V,U}}\|^{-1}_{\mathbb{L}_\infty(\lambda)}\; e^{-W^V} \leqslant \frac{d\pi_{2n+1}}{d\lambda_U} \leqslant e^{-\nu_V(V)}\|e^{\mathcal{W}_{V,U}}\|_{\mathbb{L}_\infty(\lambda)}\; e^{W^V}$$

■

Remark D.6. *The functions $(\mathcal{W}_{V,U}, \mathcal{W}^{U,V})$ defined in (D.7) can be computed explicitly for the the Gaussian model defined in (5.6) and (5.7).*

We set $\sigma_\beta = \beta\sigma\beta'$ and $\beta_\tau := \tau^{-1}\beta$. With this notation at hand, after some elementary computations (cf. for instance Exemple 6.6 in [2]) we have

$$\mathcal{W}_{V,U}(x) = \frac{1}{2}\,\mathrm{Tr}(\tau^{-1}\sigma_\beta) + \frac{1}{2}\,(\beta_\tau(x-m))'(\overline{\sigma}-\tau)\,(\beta_\tau\,(x-m)) + (\overline{m}-m_0)'\beta_\tau(x-m)$$

and

$$\mathcal{W}^{U,V}(y) = \frac{1}{2}\,\mathrm{Tr}(\tau^{-1}\overline{\sigma}) + \frac{1}{2}(y-\overline{m})'\left(\beta_\tau\,\sigma\,\beta_\tau' - \tau^{-1}\right)(y-\overline{m}) - (y-\overline{m})'\tau^{-1}(\overline{m}-m_0). \tag{D.14}$$

Note that one readily obtain the relationships

$$(D.12) \Longleftrightarrow \left(\overline{\sigma} < \tau \quad \text{and} \quad \beta\,\sigma\,\beta' < \tau\right) \Longleftrightarrow \left(\tau^{-1} < \overline{\sigma}^{-1} \quad \text{and} \quad \beta'\tau^{-1}\beta < \sigma^{-1}\right) \Longleftarrow (5.13).$$

Using (D.13), Condition $\mathcal{H}_\delta(U,V)$ ensures that for any $n \geqslant 1$ we have

$$e^{-\delta V}\left(\frac{d\pi_{2n}}{d\nu_V} \vee \frac{d\nu_V}{d\pi_{2n}}\right) \leqslant c\; e^{-\mathcal{V}_\delta} \quad \text{and} \quad e^{-\delta U}\left(\frac{d\pi_{2n+1}}{d\lambda_U} \vee \frac{d\lambda_U}{d\pi_{2n+1}}\right) \leqslant c\; e^{-\mathcal{U}_\delta}$$

for some constant $c < \infty$. These estimates immediately yield the following result.

Proposition D.7. *Under the assumptions of Proposition D.5, for any $n \geqslant 1$ we have*

$$\frac{d\pi_{2n}}{d\nu_V} \vee \frac{d\nu_V}{d\pi_{2n}} \in \mathcal{B}_\psi(\mathbb{Y}) \quad \text{and} \quad \frac{d\pi_{2n+1}}{d\lambda_U} \vee \frac{d\lambda_U}{d\pi_{2n+1}} \in \mathcal{B}_\varphi(\mathbb{X}).$$

In addition, we have the uniform estimates

$$\sup_{n\geqslant 1}\left(\|\frac{d\pi_{2n}}{d\nu_V}\|_\psi \vee \|\frac{d\nu_V}{d\pi_{2n}}\|_\psi\right) < \infty \quad \text{and} \quad \sup_{n\geqslant 1}\left(\|\frac{d\pi_{2n+1}}{d\lambda_U}\|_\varphi \vee \|\frac{d\lambda_U}{d\pi_{2n+1}}\|_\varphi\right) < \infty.$$

D.3. **Lyapunov inequalities.** The result below is a fairly direct consequence of the estimates presented in Lemma D.4.

Lemma D.8. *For any $n \geqslant 1$ we have the estimates*

$$e^{\nu_V(V)}\ e^{-(\mathcal{W}_{V,U}(x)+\mathcal{W}^{U,V}(y))} \leqslant \frac{d\delta_x\mathcal{S}_{2n}}{d\delta_x\mathcal{Q}_V}(y) \leqslant e^{-\nu_V(V)}\ e^{W^V(x)+W_U(y)}$$

and

$$e^{\nu_V(V)}\ e^{-(\mathcal{W}_{V,U}(x)+\mathcal{W}^{U,V}(y))} \leqslant \frac{d\delta_y\mathcal{S}_{2n+1}}{d\delta_y\mathcal{R}_U}(x) \leqslant e^{-\nu_V(V)}\ e^{W^V(x)+W_U(y)}.$$

Proof. Using the Sinkhorn transition operators as expressed in (2.6) together with (D.9) and (D.10), we find that, for any $n \geqslant 1$,

$$\mathcal{S}_{2n}(x,dy) \quad = \quad \frac{\mathcal{Q}_V(x,dy)e^{(V-V_{2n})(y)}}{\mathcal{Q}(e^{-V_{2n}})(x)} \geqslant e^{\nu_V(V)}e^{-(\mathcal{W}_{V,U}(x)+\mathcal{W}^{V,U}(y))}\mathcal{Q}_V(x,dy)$$

and, in the same vein,

$$\mathcal{S}_{2n+1}(y,dx) \quad = \quad \frac{\mathcal{R}_U(y,dx)\ e^{(U-U_{2n})(x)}}{\mathcal{R}(e^{-U_{2n}})(y)} \geqslant e^{\nu_V(V)}\ e^{-(\mathcal{W}_{V,U}(x)+\mathcal{W}^{U,V}(y))}\ \mathcal{R}_U(y,dx).$$

Similarly, one can resort to (D.9) and (D.10) again to obtain the lower bounds

$$\mathcal{S}_{2n}(x,dy) \quad = \quad \frac{\mathcal{Q}_V(x,dy)e^{V-V_{2n}(y)}}{\mathcal{Q}(e^{-V_{2n}})(x)} \leqslant e^{-\nu_V(V)}\ e^{W^V(x)}\ \mathcal{Q}_V(x,dy)\ e^{W_U(y)}$$

and, in the same way,

$$\mathcal{S}_{2n+1}(y,dx) \quad = \quad \frac{\mathcal{R}_U(y,dx)\ e^{(U-U_{2n})(x)}}{\mathcal{R}(e^{-U_{2n}})(y)} \leqslant e^{-\nu_V(V)}\ e^{W_U(y)}\ \mathcal{R}_U(y,dx)\ e^{W^V(x)}.$$

■

With these estimates at hand, we can now state and prove the following Lyapunov inequalities.

Proposition D.9. *Assume that Condition $\mathcal{H}_\delta(U,V)$ is satisfied for some $\delta \in]0,1]$. Then, the functions $(\varphi,\psi) = (e^{\delta U}, e^{\delta V})$ satisfy the Lyapunov inequalities*

$$\mathcal{S}_{2n}(\psi)/\varphi \leqslant c_\delta\ e^{-\mathcal{U}_\delta} \in \mathcal{B}_0(\mathbb{X}) \quad \text{and} \quad \mathcal{S}_{2n-1}(\varphi)/\psi \leqslant c_\delta\ e^{-\mathcal{V}_\delta} \in \mathcal{B}_0(\mathbb{Y}) \tag{D.15}$$

for every $n \geqslant 1$ and some finite constant $c_\delta < \infty$.

Proof. By Lemma D.8 we have

$$\begin{aligned}
\mathcal{S}_{2n}(x,dy)\ \psi(y)/\varphi(x) \quad &\leqslant \quad e^{-\nu_V(V)}\ e^{-\mathcal{U}_\delta(x)}\ \mathcal{Q}_{(1-\delta)V}(x,dy)\ e^{W_U(y)}, \quad \text{and} \\
\mathcal{S}_{2n+1}(y,dx)\ \varphi(x)/\psi(y) \quad &\leqslant \quad e^{-\nu_V(V)}\ e^{-\mathcal{V}_\delta(y)}\ \mathcal{R}_{(1-\delta)U}(y,dx)\ e^{W^V(x)},
\end{aligned}$$

which (via (D.7)) imply the inequalities

$$\begin{aligned}
\mathcal{S}_{2n}(\psi)/\varphi \quad &\leqslant \quad e^{-\nu_V(V)}\ e^{-\mathcal{U}_\delta}\ e^{\mathcal{W}_{(1-\delta)V,U}}, \quad \text{and} \\
\mathcal{S}_{2n+1}(\varphi)/\psi \quad &\leqslant \quad e^{-\nu_V(V)}\ e^{-\mathcal{V}_\delta}\ e^{\mathcal{W}^{(1-\delta)U,V}}.
\end{aligned}$$

On the other hand, we have the estimates

$$\|e^{\mathcal{W}_{(1-\delta)V,U}}\|_{\mathbb{L}_\infty(\lambda)} = \|\mathcal{Q}_{(1-\delta)V}(e^{W_U})\|_{\mathbb{L}_\infty(\lambda)} \leqslant \|e^{-W}\|_{\mathbb{L}_\infty(\lambda\otimes\nu)}\ \nu_{(1-\delta)V}\left(e^{W_U}\right)$$

and

$$\|e^{\mathcal{W}^{(1-\delta)U,V}}\|_{\mathbb{L}_\infty(\nu)} = \|\mathcal{R}_{(1-\delta)U}(e^{W^V})\|_{\mathbb{L}_\infty(\nu)} \leqslant \|e^{-W}\|_{\mathbb{L}_\infty(\lambda\otimes\nu)}\ \lambda_{(1-\delta)U}\left(e^{W^V}\right),$$

where $\nu_{(1-\delta)V}\left(e^{W_U}\right) \vee \lambda_{(1-\delta)U}\left(e^{W^V}\right) < \infty$ when Condition $\mathcal{H}_\delta(U,V)$ holds. This yields (D.15) with the constant

$$c_\delta := e^{-\nu_V(V)}\ \|e^{-W}\|_{\mathbb{L}_\infty(\lambda\otimes\nu)}\ \left(\nu_{(1-\delta)V}\left(e^{W_U}\right) \vee \lambda_{(1-\delta)U}\left(e^{W^V}\right)\right).$$

By (D.8) we have

$$\mathcal{S}_1(y,dx) \quad = \quad \frac{\mathcal{R}(y,dx)\ e^{-U(x)}}{\mathcal{R}(e^{-U})(y)} \leqslant e^{W_U(y)}\ \mathcal{R}(y,dx)\ e^{-U(x)}$$

This implies that

$$e^{-\delta V(y)}\mathcal{S}_1\left(e^{\delta U}\right)(y) \leqslant e^{-(\delta V(y)-W_U(y))}\ \|e^{-W}\|\ \lambda_{(1-\delta)U}(1)$$

Equivalently, some finite constant $c_\delta < \infty$ we have

$$\mathcal{S}_1(\varphi)/\psi \leqslant c_\delta\ e^{-\mathcal{V}_\delta}$$

This ends the proof of the proposition. ■

D.4. **Proof of Theorem 4.8.** By (C.7), the Lyapunov inequalities (D.15) in Proposition D.9 ensure that the drift conditions stated in (4.13) hold with $n_0 = 1$ and $(\varphi,\psi) = (e^{\delta(U-U_\star)}, e^{\delta(V-V_\star)})$. Recall from the proof of Proposition D.5 that $e^{-\delta U}$ and $e^{-\delta V}$ are uniformly bounded for any $\delta \geqslant 0$. Since $\mathbb{X}$ and $\mathbb{Y}$ can be exhausted respectively by the compact sub-levels sets $C_U(r)$ and $C_V(r)$ of U and V, there exists some $r_1 > 1$ such that $\lambda(C_U(r)) \wedge \nu(C_V(r)) > 0$ for $r = r_1$ and thus for any $r \geqslant r_1$. This ensures there exists some $r_0 > 0$ such that for any $r \geqslant r_0$ we have

$$\lambda(C_\varphi(r)) \wedge \nu(C_\psi(r)) > 0.$$

Following the proof of Proposition D.5, for λ-almost any $x \in C_\varphi(r)$ we have

$$\begin{aligned} e^{\mathcal{W}_{V,U}(x)} = \mathcal{Q}_V(e^{W_U})(x) \quad &= \quad \int e^{W_U(y)-W(x,y)}\ \nu_V(dy) \\ &\leqslant \quad \|e^{-W}\|_{\mathbb{L}_\infty(\lambda\otimes\nu)}\ \nu_V(e^{W_U}) \\ &\leqslant \quad \|e^{-W}\|_{\mathbb{L}_\infty(\lambda\otimes\nu)}\ \|e^{-\delta V}\|_{\mathbb{L}_\infty(\nu)}\ \nu_{(1-\delta)V}\left(e^{W_U}\right) \end{aligned}$$

and we conclude that

$$w_{V,U}(r) := \sup_{C_\varphi(r)} \mathcal{W}_{V,U} < \infty.$$

Similarly, for ν-almost any $y \in C_\psi(r)$ we have

$$\begin{aligned} e^{\mathcal{W}^{U,V}(y)} = \mathcal{R}_U(e^{W^V})(y) \quad &= \quad \int e^{-W(x,y)+W^V(x)}\ \lambda_U(dx) \\ &\leqslant \quad \|e^{-W}\|_{\mathbb{L}_\infty(\lambda\otimes\nu)}\ \lambda_U(e^{W^V}) \\ &\leqslant \quad \|e^{-W}\|_{\mathbb{L}_\infty(\lambda\otimes\nu)}\ \|e^{-\delta U}\|_{\mathbb{L}_\infty(\lambda)}\ \lambda_{(1-\delta)U}\left(e^{W^V}\right) \end{aligned}$$

which yields

$$w^{U,V}(r) := \sup_{C_\psi(r)} \mathcal{W}^{U,V} < \infty.$$

On the other hand, using Lemma D.8 for any $x \in C_\varphi(r)$ and $r \geqslant r_0$ we arrive at

$$\begin{aligned} \mathcal{S}_{2n}(x,dy) \quad &\geqslant \quad e^{\nu_V(V)}\ e^{-(W(x,y)+\mathcal{W}_{V,U}(x)+\mathcal{W}^{U,V}(y))}\ \nu_V(dy)\ 1_{C_\psi(r)}(y) \\ &\geqslant \quad e^{-w(r)}\ \nu_V(C_\psi(r))\ \mathcal{Y}_r(dy) \quad \text{with} \quad \mathcal{Y}_r(dy) := \frac{\nu_V(dy)\ 1_{C_\psi(r)}(y)}{\nu_V(C_\psi(r))} \end{aligned}$$